

CDC 2026

```
library(dplyr)

complaints <- readRDS("complaints.rds")
census_zcta_2024 <- readRDS("census_zcta_2024.rds")

api_key <- "1587e10f3f8368c0fe2579d7bae54979c1ba79a4"

url <- paste0(
  "https://api.census.gov/data/2024/acs/acs5?",
  "get=NAME,",
  "B01003_001E,",
  "B19013_001E,",
  "B17001_001E,",
  "B17001_002E,",
  "B15003_001E,",
  "B15003_022E,",
  "B15003_023E,",
  "B15003_024E,",
  "B15003_025E,",
  "B23025_003E,",
  "B23025_005E,",
  "B02001_001E,",
  "B02001_002E,",
  "B02001_003E,",
  "B02001_005E,",
  "B02001_007E,",
  "B02001_008E,",
  "B03002_001E,",
  "B03002_012E,",
  "B01002_001E",
  "&for=zip%20code%20tabulation%20area:*",
  "&key=", api_key
)

census_raw <- read.csv(
  url,
  stringsAsFactors = FALSE,
  na.strings = c("", "null", "-666666666", "-888888888")
)
```

Cleaning and Prepping Data

```
complaints_2025 <- complaints %>%
  filter(
    `Date received` >= as.Date("2025-01-01"),
    `Date received` <= as.Date("2025-12-31"),
```

```

    Product == "Credit reporting or other personal consumer reports",
    `ZIP code` != "XXXXX"
  ) %>%
  select(`Date received`, Product, `ZIP code`)

head(complaints_2025, 10)

```

A tibble: 10 × 3

	<code>`Date received`</code>	<code>Product</code>	<code>`ZIP code`</code>
	<code><date></code>	<code><chr></code>	<code><chr></code>
1	2025-01-10	Credit reporting or other personal consumer repor...	32828
2	2025-01-09	Credit reporting or other personal consumer repor...	32828
3	2025-06-03	Credit reporting or other personal consumer repor...	32828
4	2025-06-03	Credit reporting or other personal consumer repor...	32828
5	2025-06-24	Credit reporting or other personal consumer repor...	75115
6	2025-06-25	Credit reporting or other personal consumer repor...	32828
7	2025-09-05	Credit reporting or other personal consumer repor...	19136
8	2025-09-05	Credit reporting or other personal consumer repor...	19136
9	2025-09-05	Credit reporting or other personal consumer repor...	19136
10	2025-09-20	Credit reporting or other personal consumer repor...	19136

```

zip_counts <- complaints_2025 %>%
  count(`ZIP code`, sort = TRUE)

num_zips <- nrow(zip_counts)
third_size <- ceiling(num_zips / 3)

top_third <- zip_counts %>%
  slice(1:third_size)

middle_third <- zip_counts %>%
  slice((third_size + 1):(2 * third_size))

bottom_third <- zip_counts %>%
  slice((2 * third_size + 1):num_zips)

```

top_third

A tibble: 6,529 × 2

	<code>`ZIP code`</code>	<code>n</code>
	<code><chr></code>	<code><int></code>
1	30349	9917
2	33025	8811
3	33178	8688
4	77449	8567
5	60411	8546
6	33311	8520
7	30253	7894
8	35215	7355

```
9 30331      7140
10 33169      6703
# i 6,519 more rows
```

middle_third

```
# A tibble: 6,529 × 2
  `ZIP code`      n
  <chr>      <int>
1 80246      122
2 83687      122
3 85742      122
4 91784      122
5 92869      122
6 93933      122
7 96755      122
8 98109      122
9 98258      122
10 98338      122
# i 6,519 more rows
```

bottom_third

```
# A tibble: 6,527 × 2
  `ZIP code`      n
  <chr>      <int>
1 52401      14
2 53073      14
3 54476      14
4 54652      14
5 55302      14
6 55319      14
7 55375      14
8 56470      14
9 56570      14
10 60186      14
# i 6,517 more rows
```

Adding In Census Data

```
# Clean the ZCTA values
zcta_clean <- gsub("[^0-9]", "", as.character(census_raw$zip.code.tabulation.area))

# Make every ZCTA exactly 5 digits
zcta_clean <- sprintf("%05s", zcta_clean)
```

```

census_zcta_2024 <- data.frame(

  ZCTA = zcta_clean,

  population = as.numeric(census_raw$B01003_001E),

  median_household_income = as.numeric(census_raw$B19013_001E),

  poverty_percent =
    100 *
    as.numeric(census_raw$B17001_002E) /
    as.numeric(census_raw$B17001_001E),

  bachelors_or_higher_percent =
    100 *
    (
      as.numeric(census_raw$B15003_022E) +
      as.numeric(census_raw$B15003_023E) +
      as.numeric(census_raw$B15003_024E) +
      as.numeric(census_raw$B15003_025E)
    ) /
    as.numeric(census_raw$B15003_001E),

  unemployment_percent =
    100 *
    as.numeric(census_raw$B23025_005E) /
    as.numeric(census_raw$B23025_003E),

  white_percent =
    100 *
    as.numeric(census_raw$B02001_002E) /
    as.numeric(census_raw$B02001_001E),

  black_percent =
    100 *
    as.numeric(census_raw$B02001_003E) /
    as.numeric(census_raw$B02001_001E),

  asian_percent =
    100 *
    as.numeric(census_raw$B02001_005E) /
    as.numeric(census_raw$B02001_001E),

  other_race_percent =
    100 *
    as.numeric(census_raw$B02001_007E) /
    as.numeric(census_raw$B02001_001E),

  two_or_more_races_percent =
    100 *

```

```

as.numeric(census_raw$B02001_008E) /
as.numeric(census_raw$B02001_001E),

hispanic_percent =
  100 *
  as.numeric(census_raw$B03002_012E) /
  as.numeric(census_raw$B03002_001E),

  median_age = as.numeric(census_raw$B01002_001E)
)

head(census_zcta_2024, 10)

```

	ZCTA	population	median_household_income	poverty_percent
1	00601	16669	19454	59.90141
2	00602	37233	21420	46.95903
3	00603	48448	20933	44.42137
4	00606	5163	20992	54.48383
5	00610	25357	24496	42.26457
6	00611	1383	20360	39.84093
7	00612	62908	24488	39.69674
8	00616	9717	21846	43.23851
9	00617	22584	25013	41.58499
10	00622	7664	26585	44.09563

	bachelors_or_higher_percent	unemployment_percent	white_percent	black_percent
1	18.19199	20.547945	79.806827	1.4457976
2	23.29432	5.133698	27.223162	1.8021647
3	28.39045	15.354331	72.884330	3.3272787
4	13.20944	0.000000	66.589192	0.1355801
5	22.73313	6.041044	18.716725	1.8929684
6	26.22222	18.721461	61.243673	2.5307303
7	29.93979	14.342392	13.259045	1.5085522
8	23.76396	18.200349	11.464444	3.0564989
9	25.05969	10.792526	8.204924	2.7453064
10	26.11718	12.006861	57.293841	4.7755741

	asian_percent	other_race_percent	two_or_more_races_percent	hispanic_percent
1	0.10798488	8.362829	10.276561	99.31010
2	0.11817474	4.563156	66.220826	97.23901
3	0.01857662	12.169749	11.517503	98.43337
4	0.23242301	25.314739	7.728065	99.67073
5	0.00000000	9.303151	70.079268	96.44280
6	0.00000000	33.767173	2.458424	97.39696
7	0.61518408	24.766325	59.645832	99.57398
8	1.13203664	21.817433	62.529587	99.54719
9	1.29295076	22.343252	65.072618	99.14098
10	0.26096033	21.542276	15.396660	98.40814

	median_age
1	45.3
2	46.6
3	46.1

4	47.9
5	45.7
6	55.9
7	45.1
8	49.0
9	43.1
10	48.1

```
complaints_with_census <- complaints_2025 %>%
  mutate(
    ZCTA = sprintf("%05s", as.character(`ZIP code`))
  ) %>%
  left_join(
    census_zcta_2024,
    by = "ZCTA"
  )
```

```
complaints_by_zip <- complaints_2025 %>%
  mutate(
    ZCTA = sprintf("%05s", as.character(`ZIP code`))
  ) %>%
  count(ZCTA, name = "complaint_count") %>%
  left_join(
    census_zcta_2024,
    by = "ZCTA"
  ) %>%
  select(
    ZCTA,
    complaint_count,
    population,
    median_household_income,
    poverty_percent,
    bachelors_or_higher_percent,
    unemployment_percent,
    white_percent,
    black_percent,
    asian_percent,
    other_race_percent,
    two_or_more_races_percent,
    hispanic_percent,
    median_age
  ) %>%
  arrange(desc(complaint_count))

complaints_by_zip <- complaints_by_zip %>%
  mutate(
    complaints_per_1000 = (complaint_count / population) * 1000,
    complaint_level = case_when(
      complaints_per_1000 < 100 ~ 0,
      complaints_per_1000 >= 100 ~ 1
    )
  )
```

```

    )
  )

complaints_by_zip <- complaints_by_zip %>%
  select(-complaints_per_1000)

```

```

library(tidyverse)
library(broom)

logistic_model <- glm(
  complaint_level ~
    median_household_income +
    poverty_percent +
    bachelors_or_higher_percent +
    unemployment_percent +
    white_percent +
    black_percent +
    asian_percent +
    other_race_percent +
    two_or_more_races_percent +
    hispanic_percent +
    median_age,
  data = complaints_by_zip,
  family = binomial
)

tidy(logistic_model, conf.int = TRUE) |>
  knitr::kable(digits = 3)

```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2.303	1.904	-1.209	0.226	-7.554	0.792
median_household_income	0.000	0.000	-1.755	0.079	0.000	0.000
poverty_percent	-0.074	0.017	-4.402	0.000	-0.107	-0.042
bachelors_or_higher_percent	0.011	0.010	1.063	0.288	-0.010	0.030
unemployment_percent	-0.039	0.026	-1.476	0.140	-0.092	0.010
white_percent	-0.054	0.018	-3.047	0.002	-0.080	-0.002
black_percent	0.018	0.017	1.083	0.279	-0.006	0.069
asian_percent	-0.034	0.029	-1.178	0.239	-0.095	0.029
other_race_percent	-0.156	0.040	-3.889	0.000	-0.234	-0.073
two_or_more_races_percent	-0.068	0.031	-2.205	0.027	-0.124	0.001
hispanic_percent	0.059	0.019	3.111	0.002	0.019	0.093
median_age	0.049	0.014	3.385	0.001	0.020	0.076

